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Pre- and post-serial metagenomic analysis of gut microbiota as a prognostic factor in patients undergoing haematopoietic stem cell transplantation

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Summary

The human gut harbours diverse microorganisms, and gut dysbiosis has recently attracted attention because of its possible involvement in various diseases. In particular, the lack of diversity in the gut microbiota has been associated with complications of haematopoietic stem cell transplantation (HSCT), such as infections, acute graft-versus-host disease and relapse of primary disease, which lead to a poor prognosis. However, few studies have serially examined the composition of the intestinal microbiota after HSCT. In this study, we demonstrated, using next-generation sequencing of the bacterial 16S ribosomal RNA gene, combined with uniFrac distance analysis, that the intestinal microbiota of patients undergoing allogeneic HSCT substantially differed from that of healthy controls and recipients of autologous transplants. Faecal samples were obtained daily throughout the clinical course, before and after transplantation. Notably, the proportions of *Bifidobacterium* and genera categorized as butyrate-producing bacteria were significantly lower in patients with allogeneic HSCT than in healthy controls. Furthermore, among allogeneic transplant recipients, a subgroup with a preserved microbiota composition showed a benign course, whereas patients with a skewed microbiota showed a high frequency of complications and mortality after transplantation. Thus, we conclude that the stability of intestinal microbiota is critically involved in outcomes of HSCT.

Keywords: stem cell transplantation, gut microbiota, next-generation sequencing, uniFrac analysis, beta-diversity.

Allogeneic haematopoietic stem cell transplantation (allo-HSCT) is one of the most potent and curative treatments for intractable haematological disorders. In recent years, the outcomes of allo-HSCT have been improved by clinical applications of target-specific antineoplastic agents, novel antimicrobial drugs, immunosuppressants and genetic tests (Gratwohl *et al*, 2005; Gooley *et al*, 2010; Horan *et al*, 2011; Kurosawa *et al*, 2013a,b). Nonetheless, the relapse of primary disease and treatment-related toxicity still result in high mortality after allo-HSCT (D'Souza & Fretham, 2017). Thus, it is necessary to develop new therapeutic strategies to overcome these unfavourable events for a better outcome.

The human gut harbours diverse microorganisms, which play significant roles in the host's life. The microorganisms provide essential nutrients to the host, such as vitamins and short-chain fatty acids, produced by food digestion. Furthermore, the gut microbiota serves as a barrier to pathogens (Sekirov *et al*, 2010; Okumura & Takeda, 2017) and regulates the local and systemic immunity (Sekirov *et al*, 2010; Okumura & Takeda, 2016). In recent years, rapidly developing next-generation sequencing (NGS) technologies have enabled a comprehensive metagenomic analysis of the human gut microbiome via evaluation of the bacterial 16S ribosomal RNA (rRNA) gene.

16S rRNA analyses have recently revealed that gut dysbiosis is associated with the aetiology of several immunological disorders, such as rheumatoid arthritis (Scher *et al*, 2013; Zhang *et al*, 2015; Maeda *et al*, 2016), inflammatory bowel disease (Andoh *et al*, 2012; Nemoto *et al*, 2012) and type 1 diabetes mellitus (Giongo *et al*, 2011). Regarding HSCT, the abnormality in gut microbiota has been shown to lead to dissemination of infections via the bloodstream (Taur *et al*,

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2012; Montassier *et al*, 2016; Mancini *et al*, 2017), graft-versus-host disease (GVHD) (Holler *et al*, 2014; Jenq *et al*, 2015; Shono *et al*, 2016; Doki *et al*, 2017; Mancini *et al*, 2017; Han *et al*, 2018), pulmonary complications (Harris *et al*, 2016), relapses of primary diseases (Peled *et al*, 2017) and a high mortality (Taur *et al*, 2014; Mancini *et al*, 2017). Therefore, intestinal microbiota may be a promising therapeutic target for improving the prognosis of allo-HSCT.

Most previous studies have evaluated intestinal dysbiosis at only one or a very limited number of time points after HSCT by using alpha-diversity parameters, such as the Shannon–Wiener index or Simpson index. Although intestinal microbiota in each individual patient could change over the clinical course, few studies have examined fluctuations in gut microbiota or abnormalities in its bacterial composition during HSCT. Thus, in this study, we conducted 16S rRNA gene analyses using NGS to evaluate the time-dependent changes in the intestinal bacterial microbiota in patients undergoing allo-HSCT, in comparison with those in healthy controls (HCs) and recipients of autologous peripheral blood HSCT. Consequently, we determined the relationships between alterations in the intestinal microbiota composition and clinical outcomes of HSCT.

Methods

Patients

This prospective observational study was approved by the institutional review board at the Osaka University Hospital (16 198) and conducted in accordance with the Declaration of Helsinki. Written informed consent was obtained from all patients before specimen collection. A total of 24 adult patients who received allo-HSCT (n = 16) or autologous HSCT (auto-HSCT, n = 8) at the Osaka University Hospital between August 2016 and May 2017 were included in this study. Ten healthy volunteers were included in this study as an HC group. These healthy volunteers had not taken antimicrobial agents and had not been admitted to a hospital within 6 months before participating in this study.

Collection of faecal samples

Baseline faecal samples were collected at day −8 before HSCT, prior to the start of HSCT conditioning regimens. Thereafter, we sequentially collected one sample per day from the start of the conditioning regimen until 30 days after HSCT, except on days when patients could not pass stool or could not collect it by themselves due to poor physical condition. Faecal samples were put into disposable sterile faeces tubes (76 × 20 mm; Sarstedt, Nümbrecht, Germany) by the patients and then preserved in the RNAlater™ stabilization solution (Thermo Fisher Scientific, Waltham, MA, USA) at −80°C.

16S rRNA gene sequencing and data processing

Bacterial DNA was extracted from faecal samples using a DNeasy PowerSoil kit (Qiagen, Venlo, Netherlands). Each library was prepared according to the Illumina 16S Metagenomic Sequencing Library Preparation Guide with a primer set (27Fmod: 5′-AGRGTTTGATCMTGGCTCAG-3′ and 338R: 5′-TGCTGCCTCCCGTAGGAGT-3′) targeting the V1–V2 variable region of the 16S rRNA gene. A run of 251-bp paired-end sequencing of the amplicons was performed on a MiSeq instrument (Illumina, San Diego, CA, USA) using a MiSeq v2 500 cycle kit. The obtained paired-end sequences were merged using PEAR (http://sco.h-its.org/exelixis/web/software/pear/). Subsequently, 10 000 reads per sample were randomly sampled using seqtk (https://github.com/lh3/seqtk) for taxonomic assignments. The processed sequences were clustered into operational taxonomic units, with a similarity cut-off of 97%, using UCLUST version 1.2.22q. Representative sequences for each operational taxonomic unit were then classified using RDP Classifier version 2.2, with the Greengenes 13_8 database. The QIIME pipeline, version 1.9.1, was used as the bioinformatics environment for the processing of all relevant raw sequencing data (Caporaso *et al*, 2010).

Statistical analysis

All statistical analyses were performed using IBM SPSS Statistics version 19 (IBM, Armonk, NY, USA). Comparisons of the alpha diversity, unique fraction (UniFrac) distance and bacterial richness among the three groups were performed by the Kruskal–Wallis test, using the Mann–Whitney U-test with Bonferroni adjustment as a post-hoc test. A P-value of <0·05 was considered significant.

Results

Patient characteristics

The characteristics of the participants are summarized in Table I. The auto- and allo-HSCT groups consisted of four males and four females and 11 males and five females, respectively. The median age of the patients was higher in the auto-HSCT group than in the allo-HSCT group (61·5 and 46·0 years, respectively). Half of the allo-HSCT patients underwent human leucocyte antigen-mismatched transplantation. In the allo-HSCT cohort, nine patients were treated with a myeloablative regimen, whereas seven patients were treated with a reduced-intensity conditioning regimen. In the auto-HSCT cohort, all patients were treated with the myeloablative regimen. Prophylactic antibiotics were administered prior to HSCT in accordance with the clinical guidelines of the Infectious Diseases Society of America (Taplitz *et al*, 2018) or European Society of Medical Oncology (Klastersky *et al*, 2016).

Table I. Patient characteristics. Twenty-four patients were enrolled in this study; 8 were in the autologous HSCT group and 16 were in the allogeneic HSCT group. Ten healthy subjects were included in the healthy control group.

	auto-HSCT (N = 8)	allo-HSCT (N = 16)		auto-HSCT (N = 8)	allo-HSCT (N = 16)
Sex			Disease Status at transplant		
Male	4	11	CR	7	14
Female	4	5	non-CR	1	2
Age, years			Disease Risk Index		
Median	61·5	46·0	Low	NA	2
Range	(58·0–63·3)	(40·0–56·8)	Intermediate	NA	9
Primary disease			High	NA	3
AML		9	Undefined	NA	2
ALL		2	HCT-CI		
MDS		1	0	4	9
Lymphoma	4	3	1-2	3	3
MM	4		3-6	1	3
Others		1	7-	0	1
Stem cell source			Antibiotics administration 3 months		
Bone marrow		10	Carbapenems	2	4
Peripheral blood	8	4	Cephems	1	4
Cord blood		2	Polypeptides	0	1
Donor			Quinolones	3	8
Matched related		3	Antibiotics administration 1month		
Mismatched related		2	Carbapenems	0	1
Matched unrelated		3	Cephems	0	1
Mismatched unrelated		6	Polypeptides	0	0
Cord blood		2	Quinolones	0	1
Conditioning			Gut Decontamination		
Myeloablative	8	9	Levofloxacin	7	16
Reduced intensity		7	<prior 8 days	1	4
GVHD prophylaxis			>prior 7 days	6	12
CsA-based		11	None	1	0
Tac-based		5			
Pre-transplant chemotherapy					
Cytotoxic regimen	5	16			
Others	3	0			
Number of lines					
1–2	5	3			
3–4	2	5			
5-	1	8			

ALL, acute lymphocytic leukaemia; allo-HSCT, allogeneic haematopoietic stem cell transplantation; AML, acute myeloid leukaemia; auto-HSCT, autologous haematopoietic stem cell transplantation; CR, complete remission; CsA, ciclosporin; GVHD, graft-versus-host disease; HCT-CI, haematopoietic cell transplant-co-morbidity index; MDS, myelodysplastic syndrome; MM, multiple myeloma: NA, not available; Tac, tacromilus.

Comparison of pre-transplantation microbiotas

A total of 613 faecal specimens were collected (range: 3–38 samples per person; median: 28 samples). Two patients were excluded because of insufficient sampling frequency; these provided three and five samples throughout the clinical course. Of the included patients, 535 specimens were used for subsequent metagenomic sequencing analyses, whereas 70 specimens were excluded because of insufficient amounts of extracted bacterial DNA. Finally, 535 specimens from 22 patients were investigated (range: 11–34 samples per person; median: 24 samples).

First, we compared the gut microbiotas in 22 specimens obtained before conditioning therapy. The bacterial compositions at the genus level for the three groups are presented in Fig 1. Two patients from the allo-HSCT group received antimicrobial therapy at the time of specimen collection; one received levofloxacin for one month and doripenem for two weeks prior to HSCT because of febrile neutropenia, and the other received daptomycin and tazobactam/piperacillin for two weeks prior to HSCT because of catheter-related blood stream infection. The major phyla in the gut microbiotas of the two patients were *Actinobacteria* and *Firmicutes*, respectively. In contrast, the gut microbiotas of the other HSCT

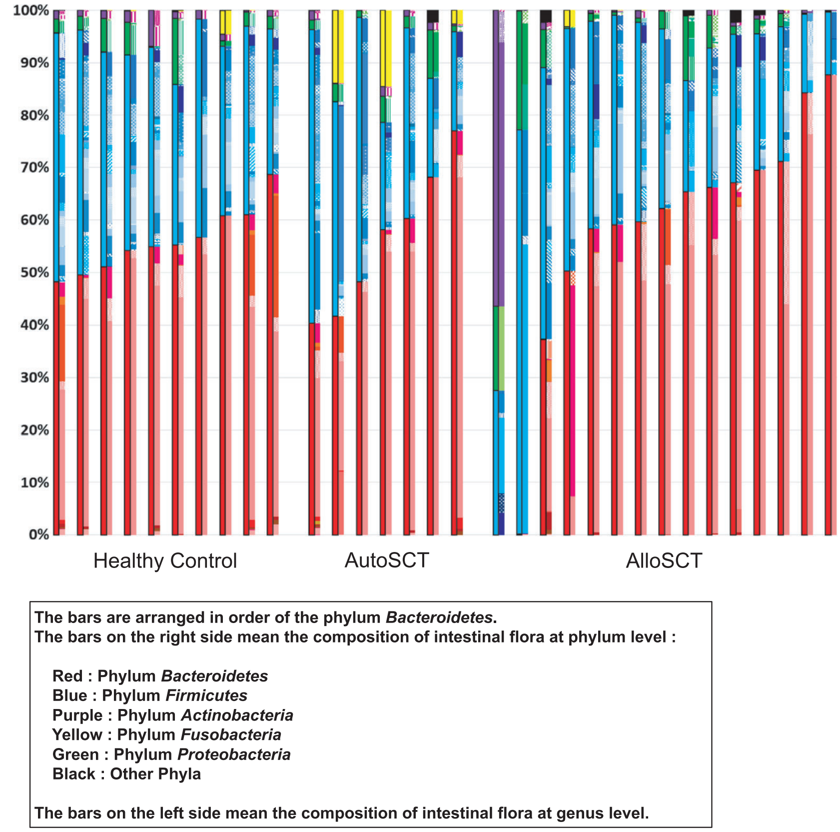

Fig 1. Intestinal microbiota compositions in initial specimens from allogeneic (allo) and autologous (auto) stem cell transplantation (SCT) patients and healthy controls. The bars on the right show the composition of intestinal flora at the phylum level; red: Phylum *Bacteroidetes*, blue: Phylum *Firmicutes*, purple: Phylum *Actinobacteria*, yellow: Phylum *Fusobacteria*, green: Phylum *Proteobacteria*, black: Other Phyla. The bars on the left show the composition of intestinal flora at the genus level.

patients and HC subjects commonly contained *Bacteroidetes* as one of the major phyla.

Next, for microbial community comparisons, we performed uniFrac distance analysis of our 16S rRNA gene sequencing data. Both unweighted and weighted uniFrac distances were calculated to evaluate the diversity of microbial components and the structure of microbiotas, respectively (Lozupone & Knight, 2005; Lozupone *et al*, 2007). The uniFrac distance analysis was used to evaluate both the diversity of microbial components and the structure of microbial flora using "Unweighted uniFrac" and "Weighted uniFrac", respectively. While "Unweighted uniFrac" does not take the number of RNA sequence reads into consideration in order to present changes in microbial components, "Weighted uniFrac" does so, to show changes in microbial flora structure. Subsequently, the analysed data were processed using uniFrac principal coordinate analysis and presented in three-dimensional coordinates (Fig 2A; Figure S1A). The weighted uniFrac distance analysis showed a higher inter-individual variability in the gut microbiota composition of the allo-HSCT recipients than in that of HCs and the auto-HSCT recipients. Interestingly, the unweighted uniFrac distance analysis also demonstrated significant differences in microbial components between allo-HSCT–HC and HC–HC ($P < 0{\cdot}001$; Figures S1B, S2). Furthermore, the weighted uniFrac distance analysis revealed significant differences in the microbial structure between auto-HSCT–HC and HC–HC ($P < 0{\cdot}001$), as well as between allo-HSCT–HC and HC–HC ($P < 0{\cdot}001$; Fig 2B, Figure S2). Thus, we concluded that there were substantial differences in the gut microbiota between healthy people and patients with haematological diseases undergoing HSCT, based on not only the diversity but also the change of components.

We also examined these comparisons using the conventional alpha diversity methods, such as Simpson's index or the Shannon-Wiener index (Fig 2C, 2D). Except the two patients on antimicrobial therapy, the diversity of the intestinal microbiota appeared to be similar in the three groups. Indeed, no significant differences in the microbial diversity of the three groups were detected by conventional methods. This failure to reveal differences in the diversity of the microbiotas between patients and HCs, despite intensive antibiotic treatment histories of the patients, suggested limitations of the conventional statistical methods used.

Differences in pre-transplant gut microbiotas

Next, we compared the relative abundances of each bacterial genus to determine differences in the gut microbiota composition among the three groups before HSCT. In the patients scheduled to undergo allo-HSCT, the abundances of *Bifidobacterium*, *Sutterella* and genera categorized as butyrate-producing bacteria, such as *Anaerostipes*, *Butyricimonas*, *Coprococcus*, *Faecalibacterium* and *Lachnospiraceae*, were

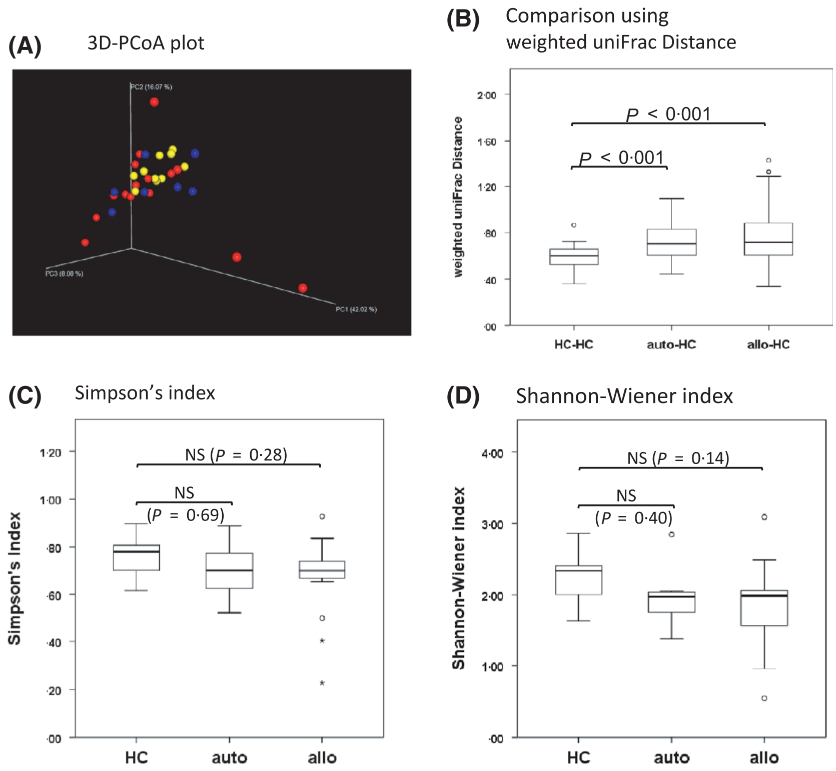

Fig 2. Comparison of the intestinal microbiota compositions using β-diversity index and α-diversity index. (A) Three-dimensional principal coordinate analysis plots based on weighted uniFrac distances. Red: allo-HSCT recipients (n = 16); blue: auto-HSCT recipients (n = 8); yellow: HCs (n = 10); (B) Weighted uniFrac distances reflecting differences in the community structure; (C) Simpson's diversity index; (D) Shannon–Wiener diversity index. NS, not significant; HC, healthy controls; auto, auto-HSCT recipients; allo, allo-HSCT recipients; NS, not significant. *P < 0·05 by Mann–Whitney U-test. allo: allogeneic haematopoietic stem cell transplantation recipient;auto: autologous haematopoietic stem cell transplantation recipient; HC, healthy controls; NS, not significant.

significantly low compared to those in the HC group (Fig 3). We also observed low relative abundances of these bacterial genera in the patients undergoing auto-HSCT, although the differences did not reach statistically significant levels, except for *Lachnospiraceae*. Notably, the abundance of *Enterococcus* was significantly higher in the allo-HSCT group.

Sequential analysis of gut microbiotas

Previous studies have reported that alteration of gut microbiota and loss of diversity frequently occur in the early period of HSCT, between the conditioning treatment and donor cell engraftment. However, it has remained unclear whether the timing and magnitude of microbiota alteration vary among patients. Thus, we daily evaluated the gut microbiota in each patient from the start of conditioning treatment until 30 days after HSCT.

Interestingly, this analysis divided the HSCT patients into two subgroups, with essentially different manifestations of the microbiota alteration, regardless of antibiotic administration or gut decontamination throughout the course of HSCT. Figure 4 shows representative cases. In some patients undergoing allo-HSCT, the microbiota composition and diversity drastically fluctuated over the course of HSCT (Fig 4A), whereas other patients with allo-HSCT maintained a robustly stable microbiota (Fig 4C). Similar results were obtained in the patients from the auto-HSCT group (Fig 4B, 4D). We could not identify the factors that might affect prognosis, such as the type of antibiotic administered, enteric decontamination, type of donor for allo-HSCT and conditioning therapy, because the sample size may be small in this cohort (Tables I and II). Therefore, we concluded that our data confirmed the existence of a subset of HSCT patients whose gut microbiota is sustainable, despite intensive chemoradiotherapy and antibiotic treatment.

Intestinal dysbiosis and clinical outcomes

Finally, we analysed whether the alteration in the microbiota composition and/or the loss of microbial diversity correlated with patient survival. To this end, we defined the alteration in the microbiota composition as a replacement of at least one of the four major bacterial taxa at pre-transplantation by other bacteria after transplantation. Additionally, the loss of diversity was defined as the case where one or two major bacterial taxa accounted for more than 90% of the whole microbiota after transplantation. Two patients were excluded from this analysis because of insufficient frequencies of sample collection due to patients' poor condition. Among the remaining 22 patients treated with HSCT, 12 patients showed altered microbiota compositions and a loss of diversity, whereas nine patients retained their microbiota characteristics and one patient showed altered flora with preserved diversity (Fig 5). Surprisingly, we found that the 2-year mortality rate was as high as 66·7% in the 12 cases with the vulnerable microbiotas, whereas all the ten patients who retained the original microbiota composition or diversity were alive for at least 2 years after HSCT (Table II, Figs 5

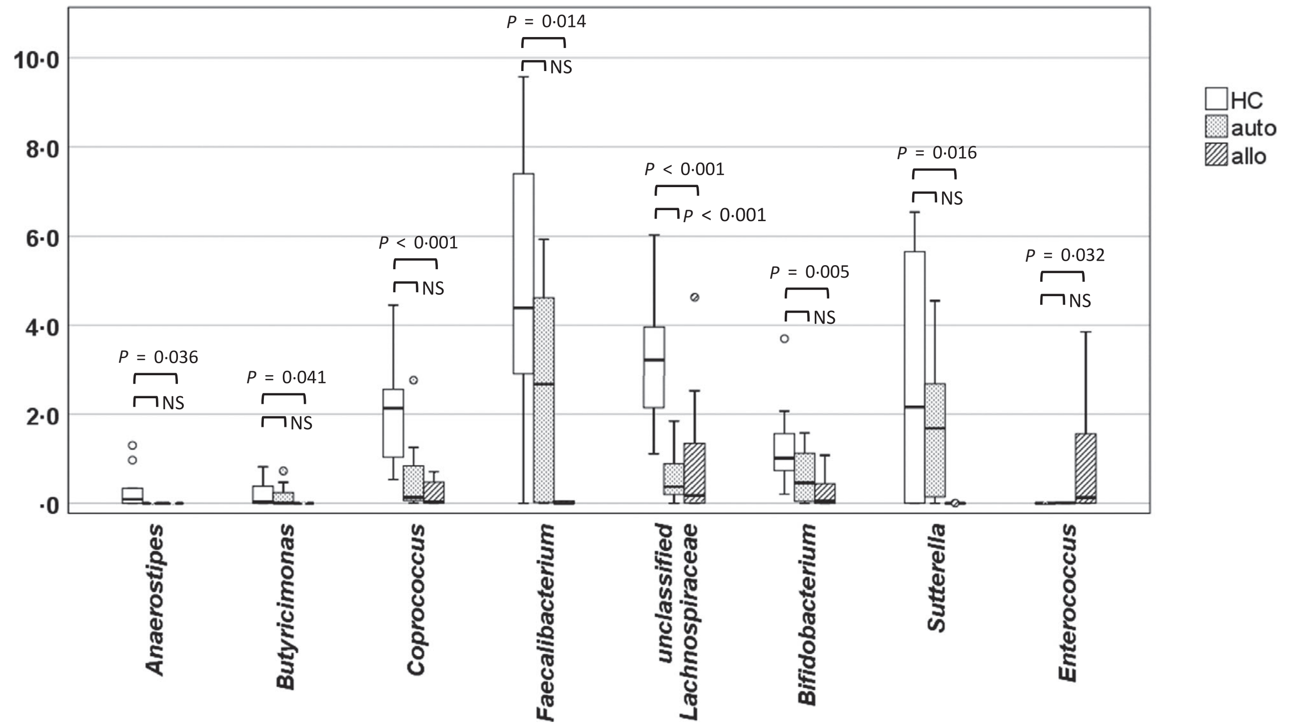

Fig 3. Relative abundances of beneficial bacterial genera in specimens from allogeneic (allo) (n = 16) and autologous (auto) HSCT patients (n = 8) and healthy controls (HCs) (n = 10). *$P < 0{\cdot}05$ by Mann–Whitney *U*-test. allo: allogeneic haematopoietic stem cell transplantation recipient; auto: autologous haematopoietic stem cell transplantation recipient; HC, healthy controls; NS, not significant.

and 6A). Additionally, the 2-year overall survival was markedly different between the two subgroups of the allo-HSCT patients (Fig 6B). We also determined that the administration of carbapenems, cephems, or polypeptides, or even gut decontamination did not significantly contribute to flora alteration and diversity loss among the three groups (Table II). Hence, we concluded that the alteration in the microbiota composition and the loss of diversity are strongly associated with patient survival after HSCT.

Discussion

The bacterial gut microbiota plays a major role in sustaining a normal physiology, and its disruption is involved in various diseases. Recently, intestinal dysbiosis, detected before or during HSCT, has been shown to correlate with detrimental consequences in the clinical course of patients (Taur *et al*, 2012; Holler *et al*, 2014; Taur *et al*, 2014; Jenq *et al*, 2015; Harris *et al*, 2016; Montassier *et al*, 2016; Shono *et al*, 2016; Doki *et al*, 2017; Mancini *et al*, 2017; Peled *et al*, 2017; Han *et al*, 2018). However, given that the conclusions were made on single or limited faecal samples, the previous studies have only provided information about alpha diversity, which describes the number and proportion of bacterial species in each individual patient at a certain time point during the long-running HSCT treatment. To learn more about the relationship between intestinal dysbiosis and treatment outcomes, it is pivotal to evaluate beta diversity, which determines dissimilarity in the microbiota among serial samples from one patient or among diverse patient groups. Therefore, we hypothesised that conventional approaches were insufficient to describe the pre-transplant dysbiosis and dynamic microbiota changes in HSCT patients. To address this issue, we conducted serial metagenomic analyses in HSCT patients.

We first assessed differences in the gut microbiota among allo-HSCT, auto-HSCT and HC groups prior to the start of conditioning regimens. We used uniFrac distance analysis to determine not only quantitative but also qualitative variations in the gut microbiota, which allowed us to clearly distinguish the microbiota characteristics in the three groups. This analytical method is useful for evaluating beta diversity, which reflects the biological diversity of the microbiota among partitioned environments (Lozupone *et al*, 2007). Using this method, Miyake *et al* (2015) successfully demonstrated the difference between the intestinal microbiota of HCs and patients with relapsing-remitting multiple sclerosis, despite a high similarity in the richness of bacterial species between the two groups).

In contrast, Simpson's index and other methods for alpha-diversity evaluation have been used for determining the distribution of microbes in a set of samples obtained from an individual subject and are not suitable for analysing samples obtained from multiple subjects. Although patients with preserved alpha diversity were reported to be classified

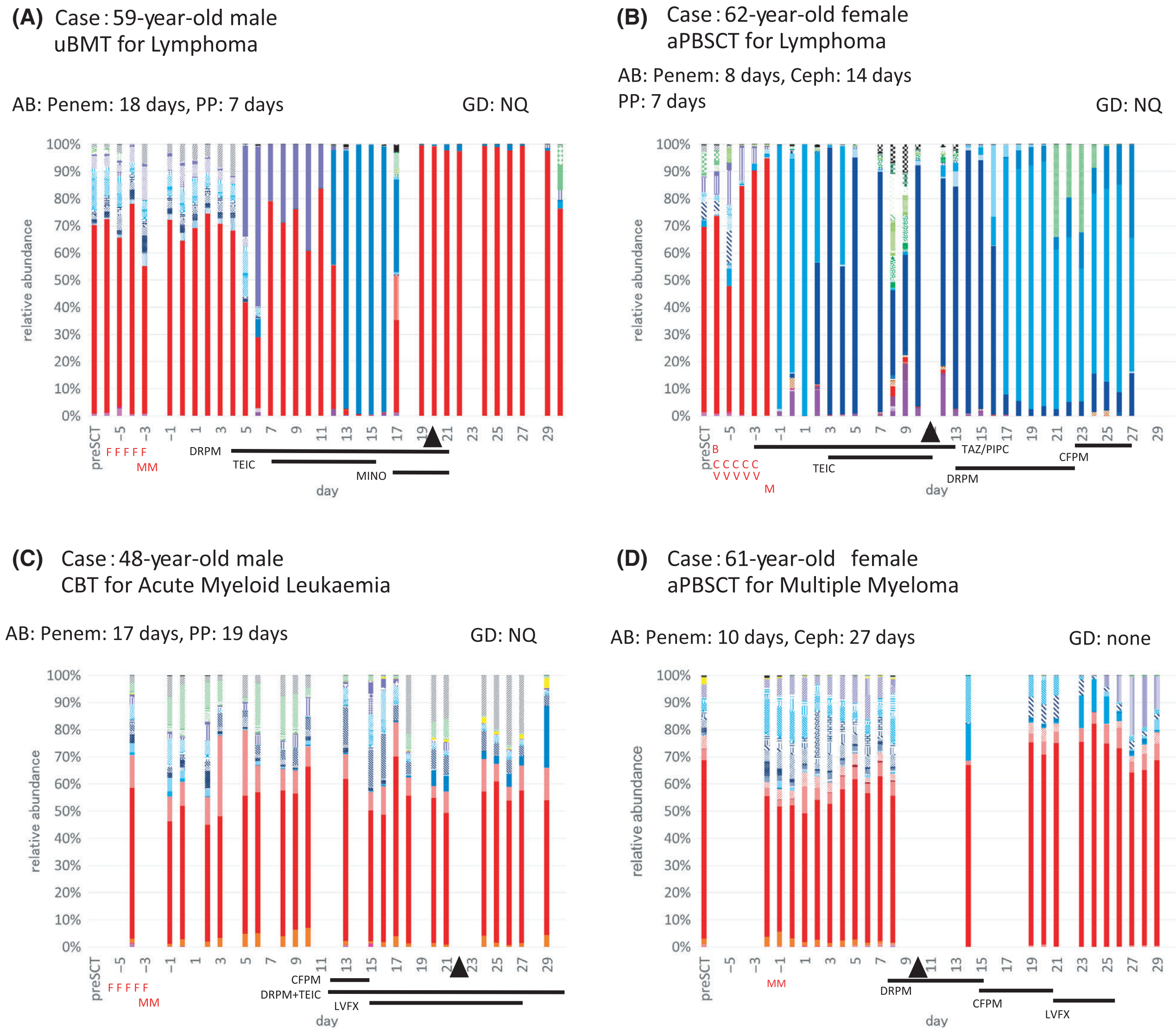

Fig 4. Typical cases of patients with or without microbiota alteration and diversity loss. (A) Allogeneic haematopoietic stem cell transplantation (HSCT) patients with microbiota alteration and diversity loss. (B) Autologous HSCT patients with microbiota alteration and diversity loss. (C) Autologous HSCT patient with microbiota alteration and diversity loss. (D) Autologous HSCT patient without microbiota alteration and diversity loss. AB, antibiotics; aPBSCT, autologous peripheral blood stem cell transplantation; CBT, cord blood transplantation; Ceph, cephems; CFPM, cefepime; DRPM, doripenem; GD, gut decontamination; LVFX, levofloxacin; MINO, minocyclin; NQ, new quinolone; Penem, carbapenems; PIPC, piperacillin; PP, polypeptides; SCT, stem cell transplantation; TAZ, tazobactam; TEIC, teicoplanin; uBMT, unrelated bone marrow transplantation, Black triangles: the day of neutrophil recovery, Red: Phylum *Bacteroidetes*, blue: Phylum *Firmicutes*, purple: Phylum *Actinobacteria*, yellow: Phylum *Fusobacteria*, green: Phylum *Proteobacteria*, black: Other Phyla. Red characters indicate; F: Fludarabine, M: Melphalan, B: BCNU, C: Cyclophosphamide, V: Etoposide.

as a good risk in previous studies, we decided that beta diversity analysis would be more suitable for analysis because it encompasses both the diversity of the microbiota and the changes of microbiota composition among multiple subjects. Our results suggested that the gut microbiota of allo-HSCT recipients substantially differed from that of HCs in terms of both community structure and community members. On the other hand, only differences in the microbial community structure could be detected between auto-HSCT recipients and HCs, probably because the allo-HSCT recipients had been more intensively treated with cytotoxic chemotherapies, immunosuppressive drugs and antimicrobial agents prior to HSCT. Indeed, the allo-HSCT patients had received, on average, 4·625 courses (range: 1–8) of cytotoxic chemotherapy, whereas the auto-HSCT patients had received only 1·875 courses (range: 0–5) before HSCT. These observations also indicate the high sensitivity of our analytical method for assessing intestinal microbiota disruption, which probably reflects the degree of intestinal damage. Using this analytical method, we believe that we can evaluate distances among

Table II. Overall survival of patients with preserved or altered microbiota composition without diversity loss

Flora alteration/Diversity loss	+/+	−/−	+/−	*P*-value
All patients (n = 22)	12	9	1	
Alive	5	9	1	*P = 0·0393
Dead	7	0	0	
Administration of antibiotics				
Carbapenems	11	9	1	NS
Cephems	12	9	1	NS
Polypeptides	11	5	0	NS
Gut decontamination	11	9	0	NS
allo-HSCT patients (n = 14)	9	5	0	
Alive	3	5		*P = 0·0269
Dead	6	0		
Administration of antibiotics				
Carbapenems	9	5		NS
Cephems	9	5		NS
Polypeptides	8	4		NS
Gut decontamination	9	5		NS
auto-HSCT patients (n = 8)	3	4	1	
Alive	2	4	1	P = 0·375
Dead	1	0	0	
Administration of antibiotics				
Carbapenems	2	2	1	NS
Cephems	3	4	1	NS
Polypeptides	3	1	0	NS
Gut decontamination	2	4	0	NS
All patients (n = 22)	12	9	1	
Neutrophil engraftment	day 12 (10–26)	day 13 (11–22)	day 10	NS
allo-HSCT patients (n = 14)	9	5	0	
Neutrophil engraftment	day 16 (11–26)	day 15 (13–22)		NS
auto-HSCT patients (n = 8)	3	4	1	
Neutrophil engraftment	day 10 (10–11)	day 11 (10–11)	day 10	NS

'Flora alteration' was defined as a replacement of at least one of the four major bacterial taxa at pre-transplantation by other bacteria after transplantation. 'Diversity loss' was defined as the case where one or two major bacterial taxa accounted for more than 90% of the whole microbiota after transplantation. One patient showed altered microbiota composition without diversity loss.

allo-HSCT, allogeneic haematopoietic stem cell transplantation; auto-HSCT, autologous haematopoietic stem cell transplantation; NS, not significant.

*P < 0·05 by logrank test.

three groups (HC/auto-HSCT/allo-HSCT) by the same yardstick.

Moreover, our data suggested that the beneficial bacterial microbiotas had already been damaged in patients before HSCT. We attributed this finding to multiple factors, such as prior cytotoxic therapies, antibiotic treatments and an immunocompromised state of the patients due to their haematological diseases. It is noteworthy that beneficial bacteria, such as *Bifidobacterium* spp., which affected butyrate-producing bacteria, were significantly reduced in the microbiota of allo-HSCT patients prior to transplantation. *Bifidobacterium* spp. reportedly repress the exacerbation of intestinal inflammation (Setoyama *et al*, 2003). Furthermore, butyrate is one of the essential substances for the maintenance of normal mucosal function and for regulation of local immune reactions by inducing regulatory T cells (Furusawa *et al*, 2013). It has been shown that the butyrogenic effects result from cross-feeding interactions between *Bifidobacteria* and butyrate-producing colon bacteria (Riviere *et al*, 2016). Moreover, this microbial metabolome is worthy of attention, and butyrate is reported to be sufficient for protection against GVHD (Mathewson *et al*, 2016; Turroni *et al*, 2016). Thus, a diminished number of *Bifidobacterium* before HSCT may be associated with unfavourable outcomes after allo-HSCT, possibly due to a high occurrence of intestinal GVHD or other complications.

Our serial evaluation of faecal samples on a daily basis throughout HSCT treatment identified patients with a poor prognosis, whose microbiotas were lacking stability and diversity. Our results partially support those of a previous

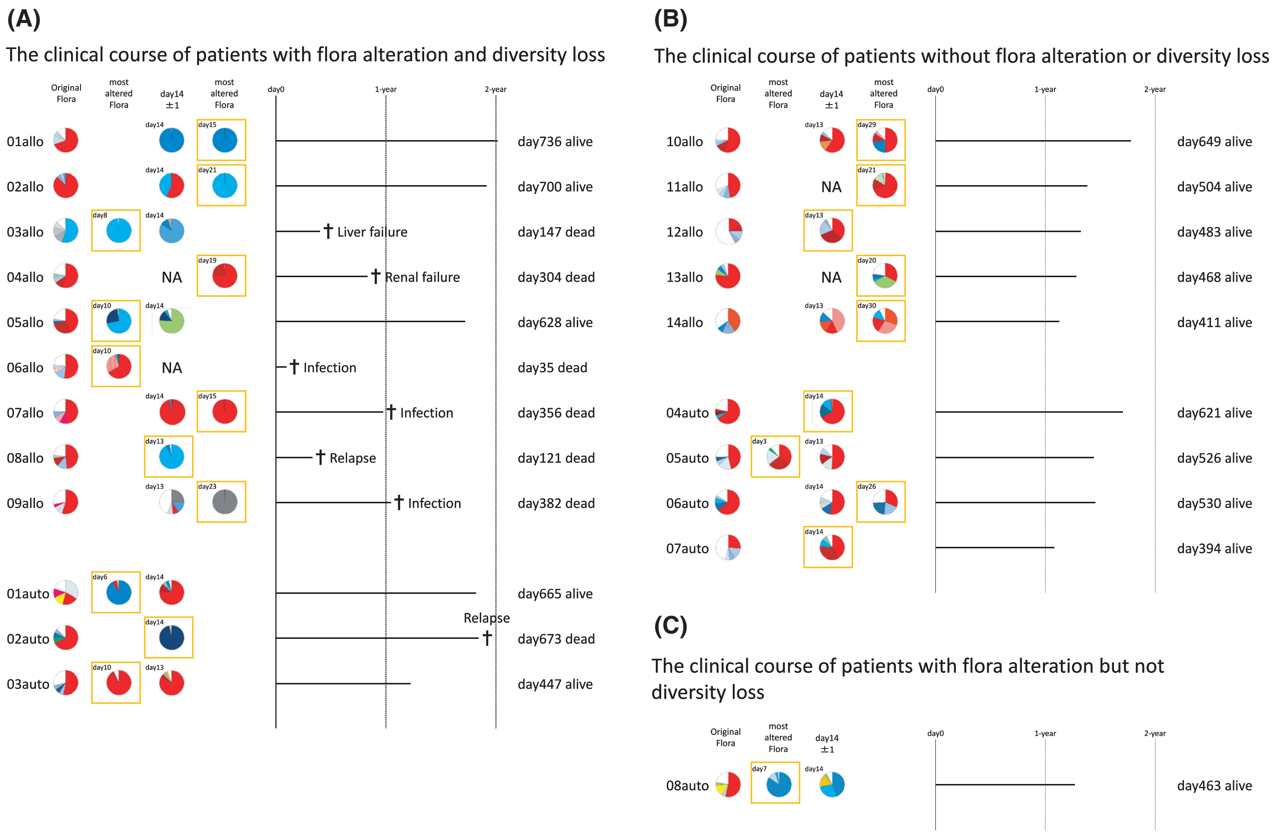

Fig 5. Clinical course of patients. From left to right, the first pie charts, pre-transplant flora; the second or fourth pie charts, most altered flora before/after day 14; the third pie charts, day14 (±1) flora. The most altered flora are indicated by orange boxes. Bars represent treatment outcome. (A) Clinical course of patients with flora alteration and diversity loss. (B) Clinical course of patients without flora alteration or diversity loss. (C) Clinical course of patients with flora alteration but preserved diversity. allo: allogeneic haematopoietic stem cell transplantation recipient; auto: autologous haematopoietic stem cell transplantation recipient; NA, not available.

report showing that flora diversity at pre-transplant stages had no significant impact on overall survival after allo-HSCT (Doki *et al*, 2017), because our data have determined the importance of flora stability alongside HSCT. Given that intestinal dysbiosis after HSCT is likely to be involved in complications after HSCT, it is important to implement countermeasures. Probiotics, such as *Bifidobacterium bifidum*, *Streptococcus faecalis* and *Clostridium butyricum* are now

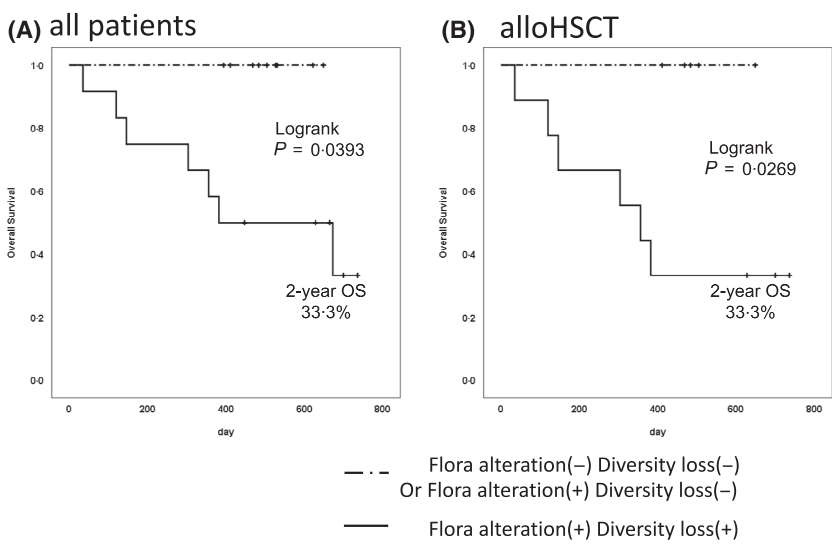

Fig 6. Two-year overall survival rate of patients with various microbiota characteristics. Dotted line, overall survival (OS) of patients with preserved flora composition and diversity (n = 9) and with altered flora composition but preserved diversity (n = 1). Solid line, overall survival of patients with altered microbiota composition and diversity loss (n = 12). alloHSCT, allogeneic haematopoietic stem cell transplantation. *P < 0·05 by logrank test.

available for clinical use, and it is worth considering the use of these probiotics during HSCT. Indeed, a recent report has shown the efficacy of synbiotics containing strains of *Bifidobacterium breve* and *Lactobacillus casei* in preventing the incidence of enteritis and pneumonia in patients with septicaemia (Shimizu *et al*, 2018). On the other hand, Suez *et al* (2018) reported that taking probiotics hindered microbiota recovery after antibiotic treatment. Randomized controlled trials are necessary to determine the positive and negative effects of probiotics on the outcomes of HSCT, but we believe that the precise choice of probiotics according to the indigenous microbiome in each individual patient (Zmora *et al*, 2018) will be promising.

As another measure to prevent dysbiosis, faecal microbiota transplantation (FMT) deserves attention. One prior study has reported that FMT was remarkably effective for refractory *Clostridium difficile*-associated colitis (van Nood *et al*, 2013). Indeed, a single FMT following an initial 4-day vancomycin administration resolved the symptoms of diarrhoea in more than 80% of patients without significant adverse events, whereas the diarrhoea resolution was only achieved in approximately 30% of patients treated with a standard vancomycin regimen for 14 days. Of note, the patients treated with FMT showed an increased microbial diversity, which was comparable to that in healthy donors. Furthermore, another pilot study has shown the safety and possible efficacy of FMT for acute intestinal GVHD in patients after HSCT (Kakihana *et al*, 2016). More recently, Taur *et al* (2018) have demonstrated the effectiveness of autologous FMT in reconstituting intestinal microbiota after antimicrobial treatment. Although these observations support the use of this unconventional approach as a promising therapy for intestinal complications after HSCT, it remains unclear whether FMT will be truly beneficial for HSCT patients under severe immunosuppressive conditions, considering potentially high risks of enteral infections with infusion via a nasoduodenal tube. Thus, further studies are needed to determine the best timing for FMT and to establish a new administration route for therapeutic microorganisms.

To our knowledge, only few studies have demonstrated, through serial microbiota analysis, fluctuations in gut microbiota of patients undergoing allo- and auto-HSCT. However, there are limitations to the present study. First, the sample cohorts were not large enough to determine the correlation of the microbiota dysbiosis with the incidence of each complication, such as intestinal GVHD or cytomegalovirus-induced enteritis, in allo-HSCT patients. Second, we were unable to estimate the absolute quantity of each microorganism in faecal samples because 16S rRNA gene analysis with NGS only determines the relative abundance of each bacterial component. Nevertheless, we found that the loss of microbial diversity predicts significantly poor survival after HSCT, and this information may be useful for making decisions regarding early intervention in these patients using unconventional methods, such as probiotics or FMT. Furthermore, we believe that future development of an analytical system, combined with serial metagenomic analyses, will provide better indicators for risks of different complications for each individual patient after HSCT.

In conclusion, the present study evaluated, in detail, the gut microbiota in patients prior to HSCT in terms of the community composition and structure. Furthermore, we elucidated the time-dependent changes in the gut microbiota of HSCT recipients by using sequential analysis, which revealed the relationship between microbiota fluctuations and patient survival after HSCT. Our results provide an important perspective on the evaluation of intestinal microbiota in HSCT patients. Future research should focus on the stabilization of the gut microbiota to improve clinical outcomes after HSCT. In addition, metabolomics analyses will enable us to determine the molecular mechanisms by which the changes in microbiota affect clinical outcomes.

Author contributions

SK, TM, HS, and KO designed the study; SK performed the statistical analysis; SK, KF, and TY contributed to manuscript preparation; DM and SN performed the metagenomic analyses; JF managed the clinical data; and TY, HS, and YK critically revised the paper. All authors approved the submitted and final version of the manuscript.

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Conflict of interests

The authors declare no conflicts of interest associated with this study.

Supporting Information

Additional supporting information may be found online in the Supporting Information section at the end of the article.

Fig S1. Unweighted uniFrac distance analysis plots distinguishing autologous (auto) and allogeneic (allo) HSCT patients and healthy controls (HCs). (A) Three-dimensional principal coordinate analysis plots based on unweighted uniFrac distances. Red, allo-HSCT recipients; blue, auto-HSCT recipients; yellow, HCs. (B) unweighted uniFrac distances, reflecting differences in community members. NS, not significant

Fig S2. Comparison of intestinal flora using weighted and unweighted uniFrac distance analysis. (A) Unweighted uniFrac distances reflecting differences in the community structure including the distance among auto-auto, auto-allo, and allo-allo; (B) Weighted uniFrac distances reflecting differences in the community structure including the distance among auto-auto, auto-allo, and allo-allo; NS, not significant; HC, healthy controls; auto, auto-HSCT recipients; allo, allo-HSCT recipients; $*P < 0{\cdot}05$, $**P < 0{\cdot}01$, $***P < 0{\cdot}005$, $****P < 0{\cdot}001$ by Mann–Whitney *U*-test.

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